D0 · Back-to-school diagnosis

S1 · 30 min · without calculator · not noted. Give a brief justification.
solved, write down what's blocking.

1. Factoring $x^2 - 5x + 6$, then resolve $x^2 - 5x + 6 \leq 0$.

2. Derivating $(x+1)e^x$ and determine the sign of the derivative.

3. $u_0 = 3$, $u_{n+1} = 2u_n$. Provide u_3 and a formula for u_n.

4. Calculate $1 + 2 + \cdots + 20$ and $1 + 2 + 4 + 8 + 16$.

5. $A(1,2)$, $B(3,5)$, $C(5,8)$. Are the points aligned?

6. $P(A) = 0{,}4$, $P_A(B) = 0{,}5$, $P_{\overline{A}}(B) = 0.2$. Calculate $P(B)$.

7. A balanced piece is launched twice independently. Calculate the probability of getting
Exactly once in a pile.

8. The statement "if $x^2 = 4$, then $x = 2$" is it true? Give the correct correction.

Corrected D0 and operation

1. $x^2 - 5x + 6 = (x-2)(x-3)$. The product is positive before 2 and after 3, negative between the two roots and no root. The whole sought is $[2,3]$.

2. By product, the derivative is $e^x + (x+1)e^x = (x+2)e^x$. Like $e^x > 0$, it is negative for $x < -2$, none in -2, positive for $x > -2$.

3. The sequence is geometrical of common ratio 2: $u_n = 3 \times 2^n$ and $u_3 = 3 \times 8 = 24$.

4. The arithmetic sum is $\frac{20(1+20)}{2} = 210$. The geometric sum is $\frac{2^5-1}{2-1} = 31$.

5. $\overrightarrow{AB} = (2,3)$ and $\overrightarrow{AC} = (4,6) = 2\overrightarrow{AB}$. Like $A \neq B$the items are aligned.

6. Total probabilities, $P(B) = 0{,}4 \times 0{,}5 + 0{,}6 \times 0{,}2 = 0{,}32$.

7. The favorable issues are pile·face and face·pile. Independence gives each the probability $\frac{1}{4}$; their meeting is likely $\frac{1}{2}$.

8. The claim is false: $x = -2$ is a counter-example. $x^2 = 4 \iff$ $x = 2$ or $x = -2$.

Resume algebra and signs if questions 1·2 fail; sequences and clues for 3·4; coordinates for 5 ; reading events for 6·7 ; logic for 8. This diagnosis is used to choose the taken over, without permanently classifying the students.